

WRO 2026 Future Engineers

Team BirTics Hardware Distribution & Selection V1

Arabic note: هذا الملف يثبت توزيع القطع واختيارها خطوة بخطوة قبل الشراء والتركيب.

Architecture	Drive	Steering	Legal target
3D printed car chassis	1 DC motor -> rear axle	1 front servo linkage	No differential drive

1. Confirmed Design Decisions

Based on the uploaded WRO 2026 Future Engineers rules, the vehicle must stay within the legal limits: 4 wheels, real driving axle, real steering actuator, max 300 x 200 mm footprint, max 300 mm height, max 1.5 kg weight, no differential wheeled base, no left/right independent drive motors for steering, no omni/ball/spherical wheels, and wired communication only during rounds.

Team BirTics Hardware Distribution V1

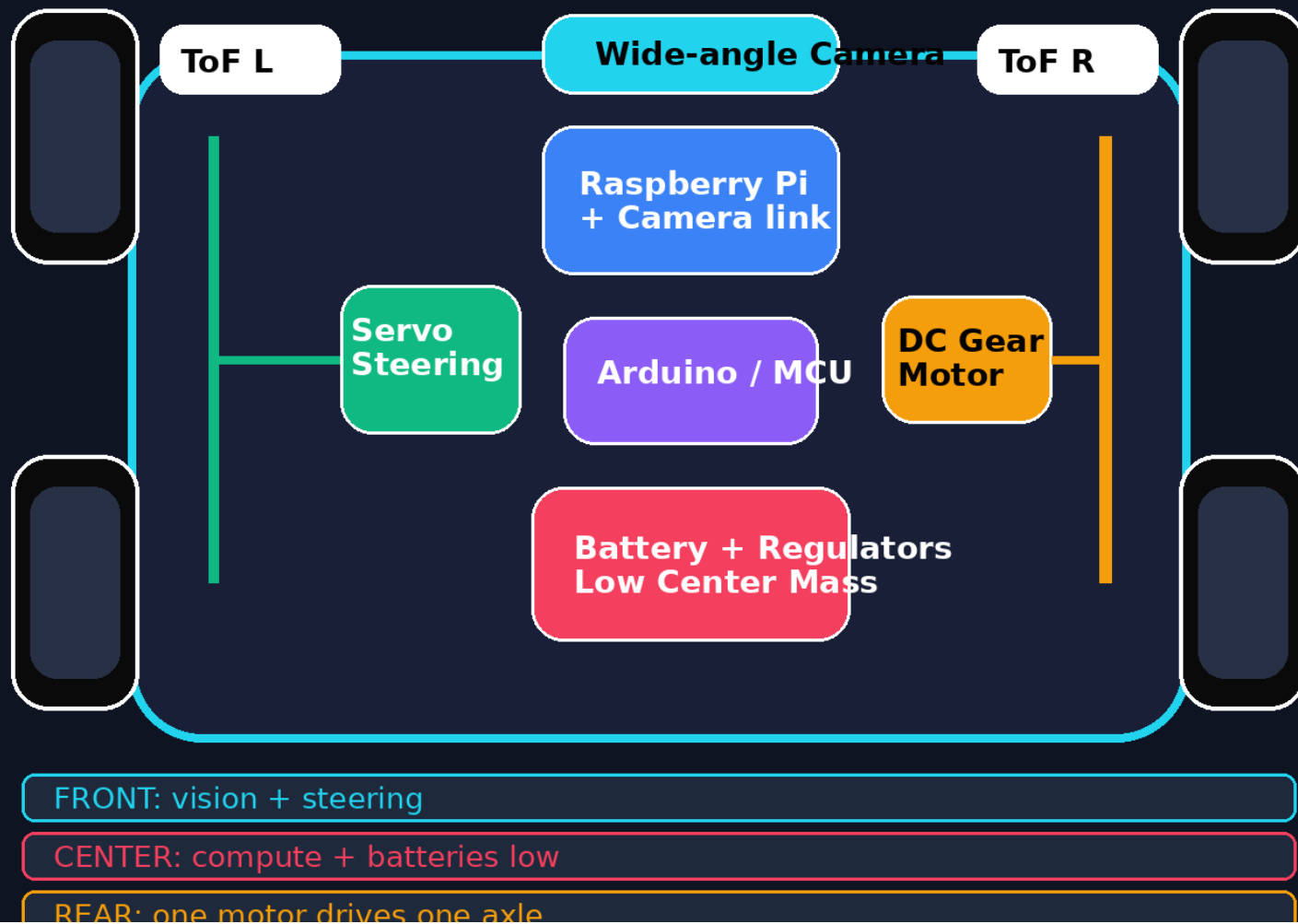


Figure 1. Proposed top-view distribution: front steering, rear axle drive, central compute and low battery placement.

2. One-Table Hardware Selection & Placement Matrix

Subsystem	Selected choice	Placement on chassis	Voltage / power target	Why we chose it	Alternative / backup	Risk / check before buying	Status
Chassis	Custom 3D printed chassis	Full base, approx. 240-270 mm length and 150-180 mm width	N/A	Maximum control over size, sensor mounts, weight distribution, and engineering documentation	Small RC chassis	Must remain under 300 x 200 mm and total height under 300 mm	SELECTED
Drive architecture	Rear-wheel drive	Rear section	Motor voltage decided by selected motor, likely 6-12V	Simple, legal, car-like, easy to document	Front-wheel drive	Do not use left motor + right motor steering	SELECTED
Driving motor	1 DC gear motor, preferably with encoder	Rear center, connected to rear axle	6-12V target; choose current after motor datasheet	One motor avoids differential-drive risk and keeps control simple	2 motors mechanically coupled to same axle	Must not independently drive left and right sides	SELECTED CONCEPT
Rear axle	Single metal rear axle with bearings	Rear, connecting both rear wheels	Mechanical part	Distributes motor motion mechanically to both wheels	Belt/pulley or gear transfer	Axle must be straight, low friction, and strong	SELECTED
Steering actuator	Metal gear servo, DS3218 preferred; MG996R backup	Front center with steering linkage	6V rail, at least 3A peak available	Real steering actuator required; strong enough for rubber wheels	MG996R for prototype	Never power servo from Raspberry Pi/Arduino 5V pin	SELECTED
Wheels	4 rubber RC wheels	Two front steering wheels, two rear driven wheels	N/A	Grip, stability, legal; supports parking and corner control	Foam/rubber wheels if same legal type	No omni, ball caster, or spherical wheels	SELECTED
Motor driver	Cytron MD10C if motor current is high; TB6612FNG only for small motors	Near rear motor, short motor wires	Input follows motor battery; logic from MCU	Reliable speed control with PWM and direction control	BTS7960 for high current	Driver current rating must exceed motor stall current	NEED MOTOR DATASHEET
Main computer	Raspberry Pi 5 or Pi 4	Center/top deck with airflow	5.1V, 5A stable supply for Pi 5	Good for camera + OpenCV processing	Jetson only if heavy AI needed	Heat, brownouts, and weight must be controlled	SELECTED
Microcontroller	Arduino Nano or UNO	Center deck near motor driver and servo signal wires	5V logic; powered from regulated 5V rail	Stable real-time control of motor, servo, encoder, button	Pico / STM32	All connections to Pi must be wired; no wireless use	SELECTED
Camera	Wide-angle camera	Front/top, adjustable tilt	Powered by Pi CSI or USB	Needed for lane, wall, red/green pillar detection	USB wide-angle camera	Mount angle must avoid glare and see near track	SELECTED
Distance sensors	2 x VL53L1X ToF sensors	Front-left and front-right	Usually 3.3V/5V module input; check board	Useful for walls, parking, obstacle confirmation	Ultrasonic sensors	ToF can be affected by angle/surface; test placement	SELECTED
IMU	BNO055 preferred	Near center of gravity	3.3V/5V module input depending board	Gives orientation; useful for turns and stability	MPU6050 lower cost	Needs calibration and vibration isolation	SELECTED
Encoder	Motor encoder or magnetic wheel encoder	On motor shaft or rear axle/wheel	3.3V or 5V depending model	Measures speed/distance, helps stop after laps and park	Optical encoder	Must be mounted securely and tested for noise	SELECTED
Power: motor	2S LiPo battery as initial target	Low center/rear-center	7.4V nominal, current depends motor	Good energy density and common for small cars	NiMH pack or 3S only if motor supports it	Need safe charger and fuse/protection plan	SELECTED TARGET
Power: Pi	Dedicated buck regulator or power bank	Low center, separate from motor noise	5.1V / 5A for Raspberry Pi 5	Prevents resets when motor/servo draw current	Official Pi supply for bench testing	Common ground required with MCU when signals shared	SELECTED
Power: servo	Dedicated buck regulator	Near servo/front center	6V, 3A+ recommended	Servo peaks can reset electronics if shared badly	5V high-current rail if servo supports it	Do not share weak Pi/Arduino supply	SELECTED
Start system	One power switch + one start push button	Power switch accessible; start button top side	Depends circuit; button to MCU GPIO	Matches required start procedure and avoids illegal interactions	Integrated button on controller	Only one start button; no switches used to enter data	SELECTED
Wiring/structure	JST/XT connectors, screw terminals, zip ties, heat shrink	All around, fixed and strain-relieved	Wire gauge based on current	Reliability and easy maintenance	Dupont only for low-current signals	Loose wires can fail during rounds	SELECTED

3. Build Order: What to Buy and Test First

Phase	Buy / prepare	Goal	Pass condition	Do not move forward until
Phase 1 - Mechanical skeleton	3D printed base, 4 rubber wheels, rear axle, bearings, steering linkage prototype	Check size, wheel alignment, and steering freedom	Car rolls smoothly by hand and steering moves without binding	Footprint and height are safely below WRO limits
Phase 2 - Motion control	DC gear motor, motor driver, steering servo, Arduino, 6V servo regulator, motor battery	Make the car drive forward/backward and steer left/right	Start button works; motor PWM and servo angles are controllable	No overheating, no servo jitter, no illegal differential behavior
Phase 3 - Vision and sensors	Raspberry Pi, wide camera, 2 ToF sensors, IMU, encoder	Add perception and feedback	Camera sees track; ToF readings stable; encoder counts correctly	Power rails remain stable while motor and servo run
Phase 4 - Competition reliability	Cable management, spare connectors, fuse/protection, final mounts	Make the robot robust for repeated rounds	10+ test runs without loose wires, resets, or mechanical failures	Documentation photos, wiring diagram, and test logs are updated

4. Final Architecture Statement for the Engineering Journal

Selected hardware architecture: Team BirTics will build a custom 3D printed four-wheel self-driving car using rear-wheel drive and front-wheel steering. A single DC gear motor drives one rear axle mechanically connected to both rear wheels. A metal gear servo controls the front steering linkage. This choice avoids differential drive, keeps the robot car-like, supports autonomous parking, and makes the design easier to reproduce and document.

5. Quick Legal Red Flags Before Buying

Do NOT buy: 2WD smart robot chassis, 4WD differential robot kit, tank/tracked chassis, omni wheels, ball caster, spherical caster, left/right independent drive motor system, or any module intended for wireless control during the round.